

Risk Neutral Default Probability

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Abstract

We present a measure change for credit default probabilities using Wang transform, a generalized Esscher transform. We show that the risk neutral default probability is simply a Wang transform of the nature default probability from the perspective of single names.

Key Words: Risk neutral default probability, Esscher transform, Wang transform

Central to the celebrated Nobel-winning idea of Black-Scholes formula is a dynamic and risk-free replication strategy of an option using stock and bond. Black and Scholes obtains a partial differential equation for the option price. Using Feynman-Kac theorem the solution to the partial differential equation can be expressed as the risk-free discounted expected value of option payoff function with respect to a *risk neutral distribution*. If the market is complete then the risk-neutral measure is unique, and the option price is unique as well. Under the assumption of Geometrical Brownian motion for the asset price the risk neutral distribution is obtained by simply changing the expected asset return to the risk-free return.

However, little has been written on the risk neutral default probability in credit analysis. Part of the reason is that credit risk is hardly replicable. For example, it is difficult to find a replication portfolio for the first-to-default contract. Also the credit market is not necessarily complete. Even though we find a risk neutral

measure it might still be not unique. The asset return under the risk neutral distribution is usually the risk-free return, but we do not know how the risk neutral default probabilities are related to their nature default probabilities.

Here we use the Merton (1974) model and Esscher transform to show how we make measure changes for default probabilities. Esscher transform has been widely used to do measure change recently. For an exemplary application of the Esscher transform we refer readers to Shiu and Gerber (1996). We can describe a credit default using a stochastic process for its default intensity or hazard rate. For detailed risk measure change using intensity model we refer interested readers to Schönbucher (2003). KMV has tried to establish a relationship between a company's expected default frequency (EDF) and its CDS spreads. This might be the first attempt to create the link between the risk neutral measure and the nature measure with respect to the default probabilities.

We first introduce some basic notations. Let us consider a general framework on a one-period setting over a time horizon $[0, T]$. We consider a single credit and use τ to denote its default time, which measures the time from today to the time that default occurs. The survival time density function and distribution functions are $f(t)$ and $F(t)$ respectively. We often call the distribution of the survival time as a *credit curve*. Remember here we use the actual probability of defaults that we observe based on the historical information, which has been done by rating agency. Or in a simple language these probabilities are probability under a nature measure of P .

The first structural model approach to assess credit risk was introduced by Merton (1974). In his model, the firm's asset value V_t is assumed to be a geometrical Brownian motion (GBM) under the P measure, starting from a known constant $V(0)$ at time zero. That is

$$\frac{dV(t)}{V(t)} = \mu dt + \sigma dW_t \quad (1)$$

where μ is the mean rate of return on asset, σ is the asset volatility, and W_t is Wiener process. In this model we use a key concept called *the distance of default*, which is the number of standard deviation by which the asset value of a firm exceeds its liability or a single bond with face value D maturing at time T . The distance of default is defined as follows:

$$X(t) = \frac{\log[V(t)] - \log(D)}{\sigma} \quad (2)$$

We note that $X(t)$ is a Brownian motion with a unit variance parameter and a constant drift of $m = (\mu - \frac{\sigma^2}{2})/\sigma$. Under the Merton model the default occurs at

the maturity date T if

$$q(T) = P[X(T) \leq 0] = \Phi[B(T)] \quad (3)$$

where

$$B(T) = \frac{X(0) + mT}{\sqrt{T}} \quad (4)$$

This style of model has been shown to have predictive power for rating migrations and default by KMV and some other researchers (Delianedis and Geske (1998)).

Next we use Esscher transform to show how to get a risk neutral distribution of asset return and the relationship between the risk neutral survival time distribution and the natural survival time distribution. Recall that the asset return under the assumption (1) can be expressed as follow:

$$V(t) = V(0)e^{(\mu - \sigma^2/2)t + \sigma W(t)} = V(0)e^{Y(t)} \quad (5)$$

We assume that the moment generating function for $Y(t)$ exists and is defined as follows:

$$M(h, t) = E[e^{hY(t)}].$$

We also have

$$M(h, t) = M(h, 1)^t,$$

which is valid for a broad category of stochastic processes, such as Lévy process and Wiener process. Then the Esscher transform process

$$\frac{e^{hY(t)}}{M(h, t)}$$

is a positive martingale, and can be used to define a change of probability measure. The risk neutral Esscher transform is the Esscher transform of parameter $h = h^*$ such that the process

$$e^{-rt}V(t)$$

is a martingale with respect to new measure under Esscher transform. Any expectation taken under the new measure by Esscher transform is denoted as $E(\bullet; h^*)$. Hence we have a condition

$$E[e^{-rt}V(t); h^*] = e^{r \cdot 0}V(0) = V(0)$$

or

$$E \left[e^{-rt} V(0) e^{Y(t)} \frac{e^{hY(t)}}{M(h, t)} \right] = V(0).$$

This can be simplified to the following

$$e^r = \frac{M(1 + h^*, 1)}{M(h^*, 1)}. \quad (6)$$

Since $Y(t)$ is a normal distribution with mean $(\mu - \sigma^2/2)t$ and variance $\sigma^2 t$, its moment generating function is as follow

$$M(h, t) = e^{(\mu - \sigma^2/2)t \cdot h + \frac{1}{2}\sigma^2 \cdot t \cdot h^2}$$

Then, it is easy to show from above condition (6) we obtain $h^* = \frac{r - \mu}{\sigma^2}$. The resulting expected asset growth rate is $\mu^* = \mu + \sigma^2 \frac{r - \mu}{\sigma^2} = r$. Hence the risk neutral default probability using the Merton model is the same as equation (3) with the change of m to $m^* = (r - \frac{\sigma^2}{2})/\sigma$. That is,

$$q^*(T) = P[X^*(T) \leq 0] = \Phi[B^*(T)] \quad (7)$$

where

$$B^*(T) = \frac{X(0) + m^* T}{\sqrt{T}} \quad (8)$$

From (3) and (7), the relationship between the risk neutral and nature or physical default probabilities is give as follows:

$$q^*(T) = \Phi \left(\Phi^{-1}[q(T)] + \frac{\mu - r}{\sigma} \sqrt{T} \right).$$

So the general relationship between the CDF functions under two measures is as follow:

$$F^*(t) = \Phi \left(\Phi^{-1}[F(t)] + \frac{\mu - r}{\sigma} \sqrt{t} \right). \quad (9)$$

We provide a simple example to show the relationship between the nature and risk neutral default probability. Suppose we have the following assumptions: $\mu = 10\%$, $r = 5\%$, $\sigma = 15\%$, and the nature hazard rate is 3% , then we have the following graph for the risk neutral and nature default probabilities. It can be seen from the Figure (1) that $F^*(0) = F(0) = 0$, $F^*(+\infty) = F(+\infty) = 1$ and $F^*(t) > F(t)$ for $0 < t < +\infty$. The measure change is associated with the risk premium $\frac{\mu - r}{\sigma} \sqrt{t}$.

This kind of transform is now called *Wang transform*, a broad measure change or transform introduced by Wang (2000). The intuitive idea of the Wang transform is very similar to the Gaussian copula model for credit portfolio modeling introduced by Li (2000). For any risk whose distribution $F(x)$ is not normal, we first use its own distribution function to translate the risk into a uniform distribution $F(X)$, then we map this uniform distribution into a normal variable by using the normal inverse function $\Phi^{-1}(F(x))$, and apply a measure change to this normal variable by shifting its mean $\Phi^{-1}(F(x)) + \lambda$, and finally use normal CDF to transform the result back into a CDF $\Phi(\Phi^{-1}(F(x)) + \lambda)$.

We conclude that from nature risk measure of default probabilities to risk neutral default probabilities we simply need to make a Wang transform with parameter h in the transform set as the *Sharpe ratio*, $\eta = \frac{\mu - r}{\sigma}$, a measure of the risk premium associated with uncertainty about the firm's asset value. One idea is to use CAPM and set $\eta = \rho\eta_M$ where ρ is the correlation between the firm's asset return with the market return and η_M is the Sharpe ration for the market portfolio, whose proxy is usually taken to be a major market index. This idea has been used by J. P. Morgan to value credit derivatives using their CreditGrade from Merton model (McNulty and Levin (2000)). Similar idea has been used by KMV (Vasicek (1995)) to generate option adjusted spread from KMV's EDF (expected default frequency). This shows that we can use Wang or Esscher transform to do measure change for the asset value as well as the default probability from a nature measure to a risk neutral measure. We have to emphasize that this risk neutral measure is not necessarily unique.

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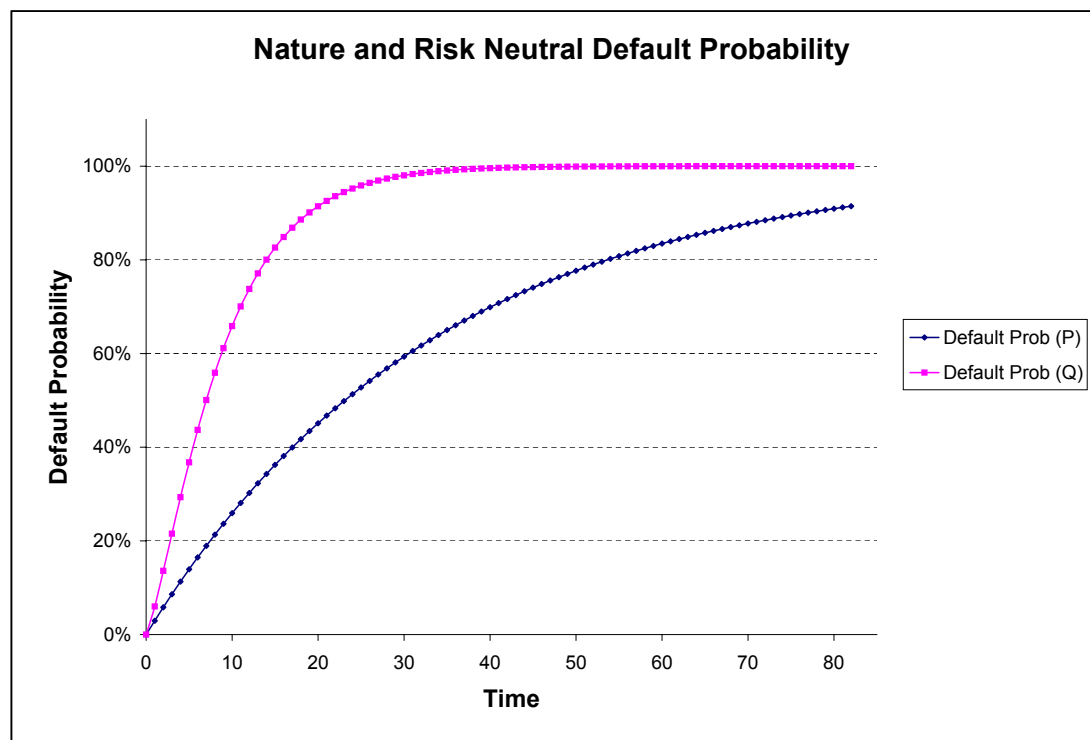


Figure 1: Nature and Risk Neutral Default Probabilities